

The research process: each surviving explanation handed to a stricter test

**Initial hypothesis:
times-through-order penalty**



✗ naive backtest: at or below breakeven

Velocity-decline signal



✗ debiasing: AUC 0.61 to 0.52 (selection)

Bullpen-fatigue multiplier



✗ no effect: fatigued pens concede no more

**Remaining-runs model
+ fatigue terms**



✗ MAE unchanged; game state already suffices

**Forecast encompassing
(joint + per-feature)**



✗ book error unpredictable, OOS R^2 near 0

**Transfer function
(ΔBook vs ΔRE)**



→ one common slope near 0.74, priced about right

Boundary identified

→ open frontier: live market microstructure